

Unity Audio

CCS16 – Game Console Integration



Agenda

- Unity Audio and Sound Effects
- Audio Mixing
- Audio Plug-ins
- Audio Playlists





Unity Audio and Sound Effects

Unity Audio



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The Importance of Audio and Sound Effects

- Sound design in a game helps with setting the theme and atmosphere of various scenes.
- **Soundtracks** are the music and ambient sounds that you hear in a game.
- **Sound effects** are all non-musical audio that is triggered through gameplay elements.



The Importance of Audio and Sound Effects

Soundtracks

- Theme songs
- Background music
- Menu music
- Combat music

Sound Effects

- Running sounds
- Interaction sounds
- Weapon swings
- Gear sounds
- Vehicle Sounds
- Voiceovers
- UI/UX sounds



The Importance of Audio and Sound Effects

- Soundtracks and sound effects are either typically a mix of in-house developments and third-party sources.
- The type of game you create will determine what type of audio and sound effects you will use.
- Some games prefer a focus on soundtracks while others place heavy emphasis on sound effects.



The Importance of Audio and Sound Effects

- There are five key elements in designing the sound for a game.
 - *Source*
 - *Envelope*
 - *Pitch*
 - *Frequency*
 - *Layering*



Elements in Sound Design

Source

- This refers to the source or inspiration of your sound effects.
- These can be taught of as the foundation of the various audio elements in your game.
- Typically, you want to record real life sounds as a basis for sound effects and then edit them later on.



Elements in Sound Design

Source

- You can also make use of existing sound libraries to help you with creating the sound profile for your game.
 - Blipsounds:
<https://blipsounds.com/community-library/>
 - Andrew V Scott:
<https://www.andrewvscott.com/Building-A-Large-SFX-Library-for-Free>
 - Freesound.org:
<https://freesound.org/>



Elements in Sound Design

Envelopes

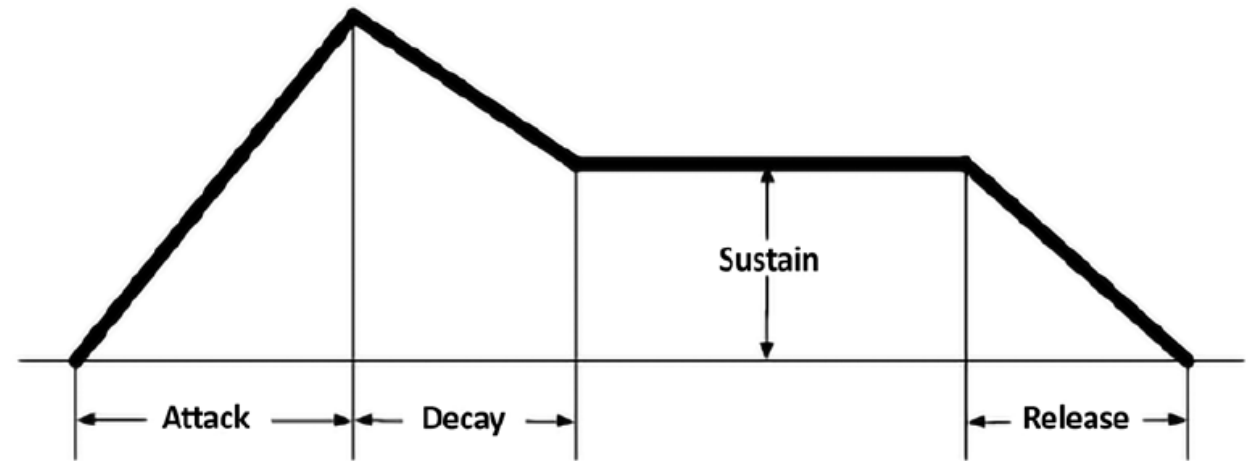
- This describes the volume of sounds effects over time.
- The two parts of an envelope are **attack** and **release**.
- These parts are typically described by their speed.



Elements in Sound Design

Envelopes

- **Attack** refers to how fast the sound volume peaks.
- **Release** refers to how the loudness of the sound and how the sound ends.



Elements in Sound Design

Pitch

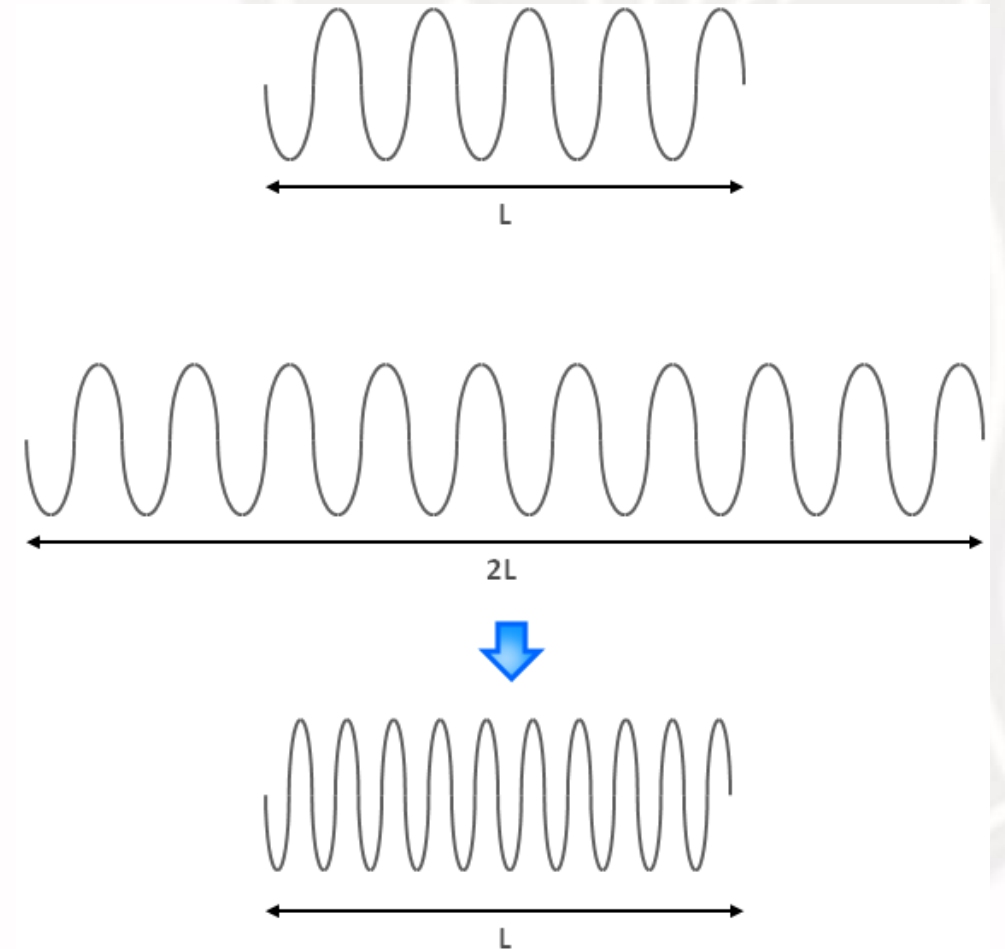
- This refers to the highness or lowness of the pitch.
- Changing the pitch of a sound will give a “wavering” effect if taken to the extreme.
- This is typically tied to the frequency of the sound.



Elements in Sound Design

Pitch

- Large characters for example will typically have low-pitched voices, while small characters will have high pitched voices.
- Pitch will also change in relation to speed, such as a vehicle speeding up or slowing down.



Elements in Sound Design

Frequency

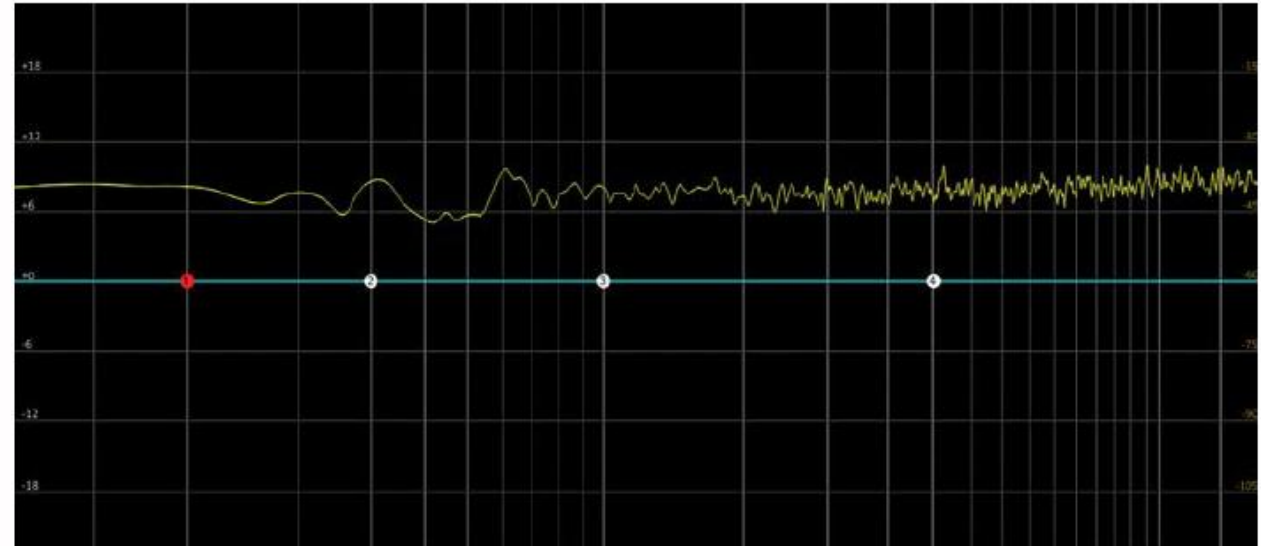
- This refers to the rate of vibration for your sound effects.
- Higher frequencies are typically referred to as the treble.
- Lower frequencies are referred to as the bass.



Elements in Sound Design

Frequency

- Using an equalizer is a way to visualize frequencies and adjust them as needed for your game.
- Adjusting frequencies will allow sound to either stand out or blend into the background.



Elements in Sound Design

Layering

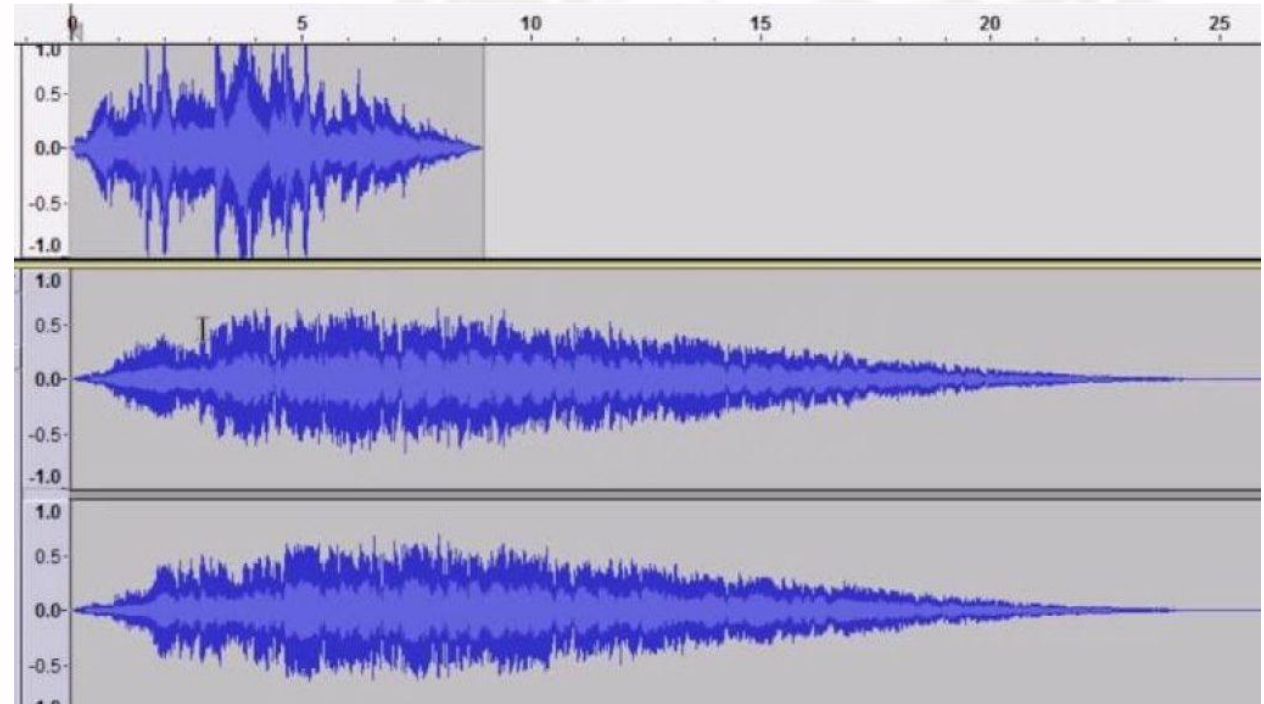
- This refers to the process of playing multiple sounds together in a unique sequence.
- This is like the concept of mixing, which is combining multiple tracks to create new music.
- Compared to mixing, layering is simply stacking different sound on top of each other.



Elements in Sound Design

Layering

- This allows you to combine two basic sound effects and create entire new sounds, such as a sword and fire, to create an enchanted fire sword.



Unity Audio Mixing

- Unity is compatible with the common types of the audio files:
 - *Audio Interchange File Format (AIFF)*
 - *Waveform Audio File Format (WAV)*
 - *MPEG Audio Layer-3 (MP3)*
 - *Ogg Vorbis (OGG)*
- Unity also supports tracker modules, which is creating instruments from audio samples.



Unity Audio Mixing

- Unity makes use of two parts when implementing audio in a game.
- The first is the **Audio Source**, which dictates where sounds will be emitted from.
- The second is the **Audio Listener**, which picks up sound effects from different sources and triggers a response.



Unity Audio Mixing

- You can also make use of **Audio Filters** to add effects to the various sounds in your game.
- These are added as additional components to a GameObject.
- These are then applied to the attached audio file when the game object is triggered or enabled for a scene.



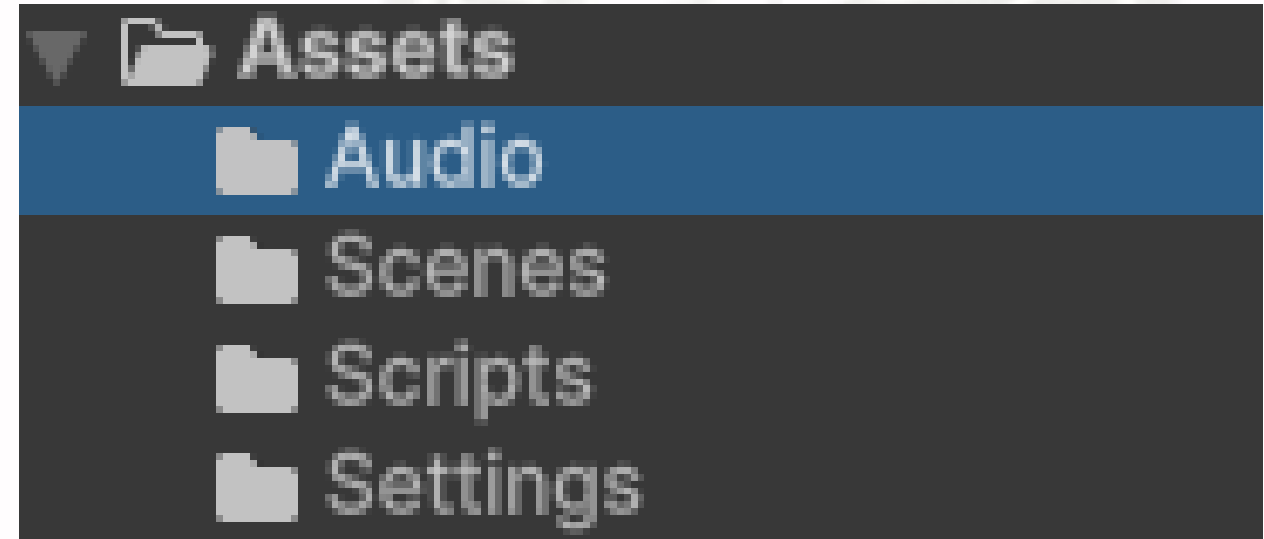
Adding Audio in Unity

- Adding audio in Unity is very simple.
- First, we will be opening or creating a project to use.
- The project should already have a GameObject that we can use as our Audio Source.



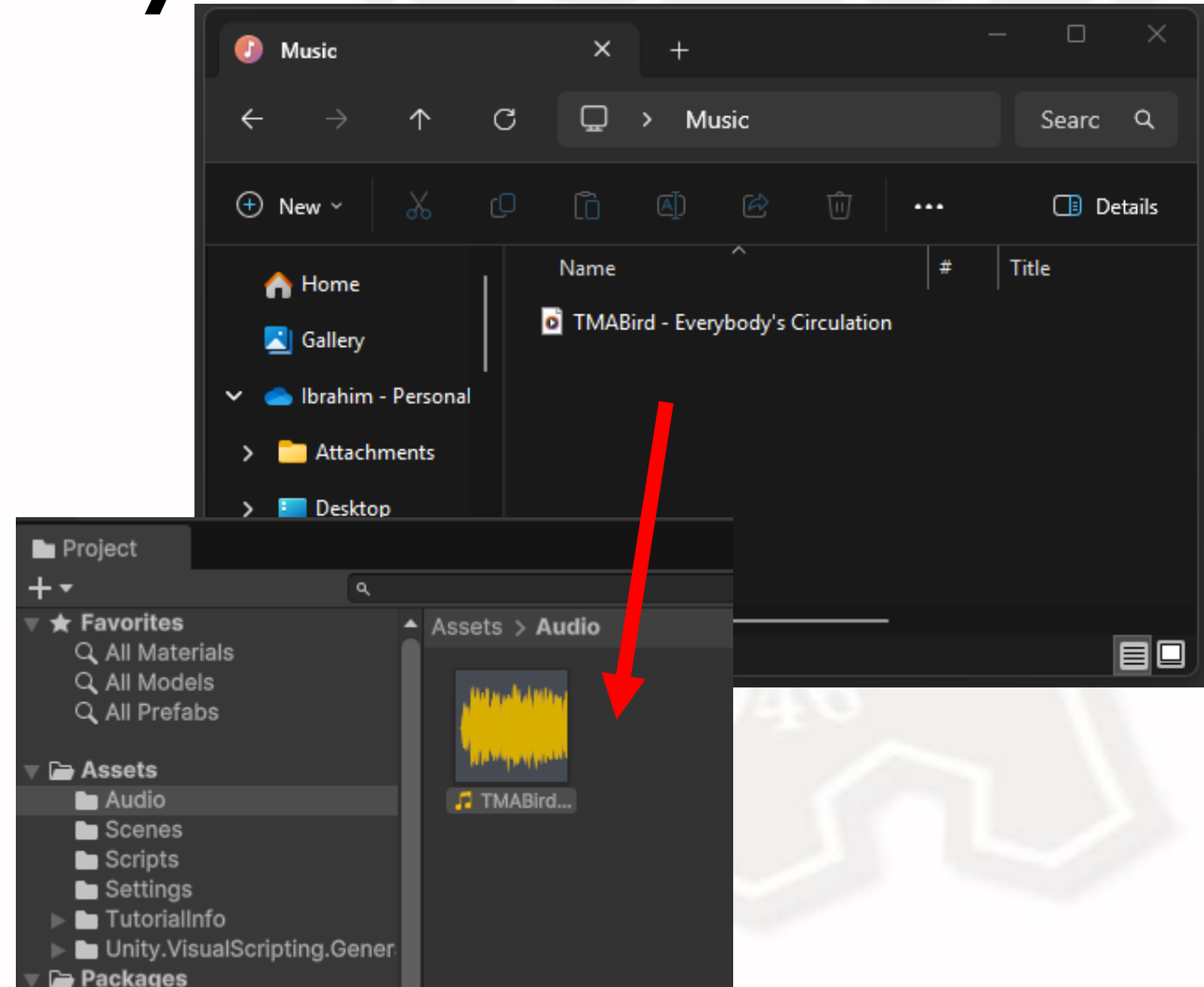
Adding Audio in Unity

- Start by creating an “Audio” Folder in your project assets.
- This is where you will be adding your game audio.



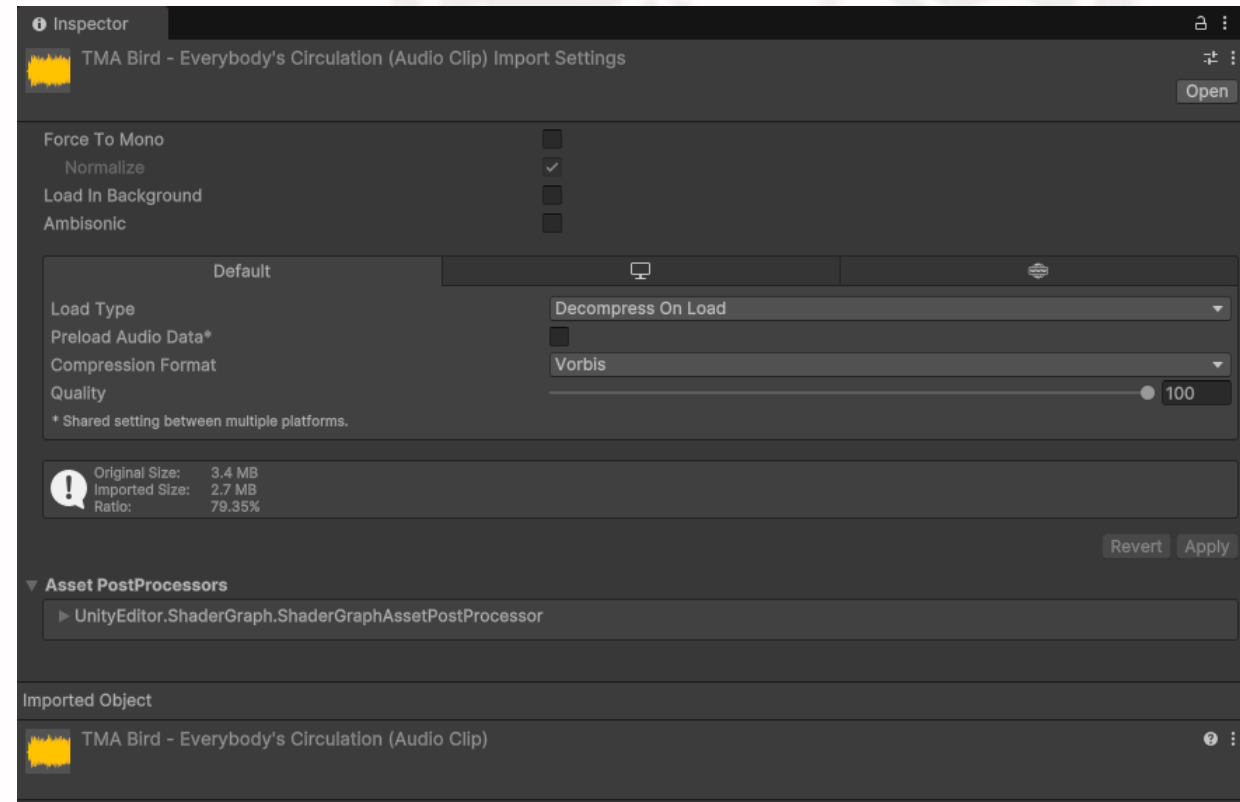
Adding Audio in Unity

- To add our audio file/s to our project, we can click and drag the files to our audio folder.
- Wait for a minute or two, and the audio will appear inside the assets folder in Unity.



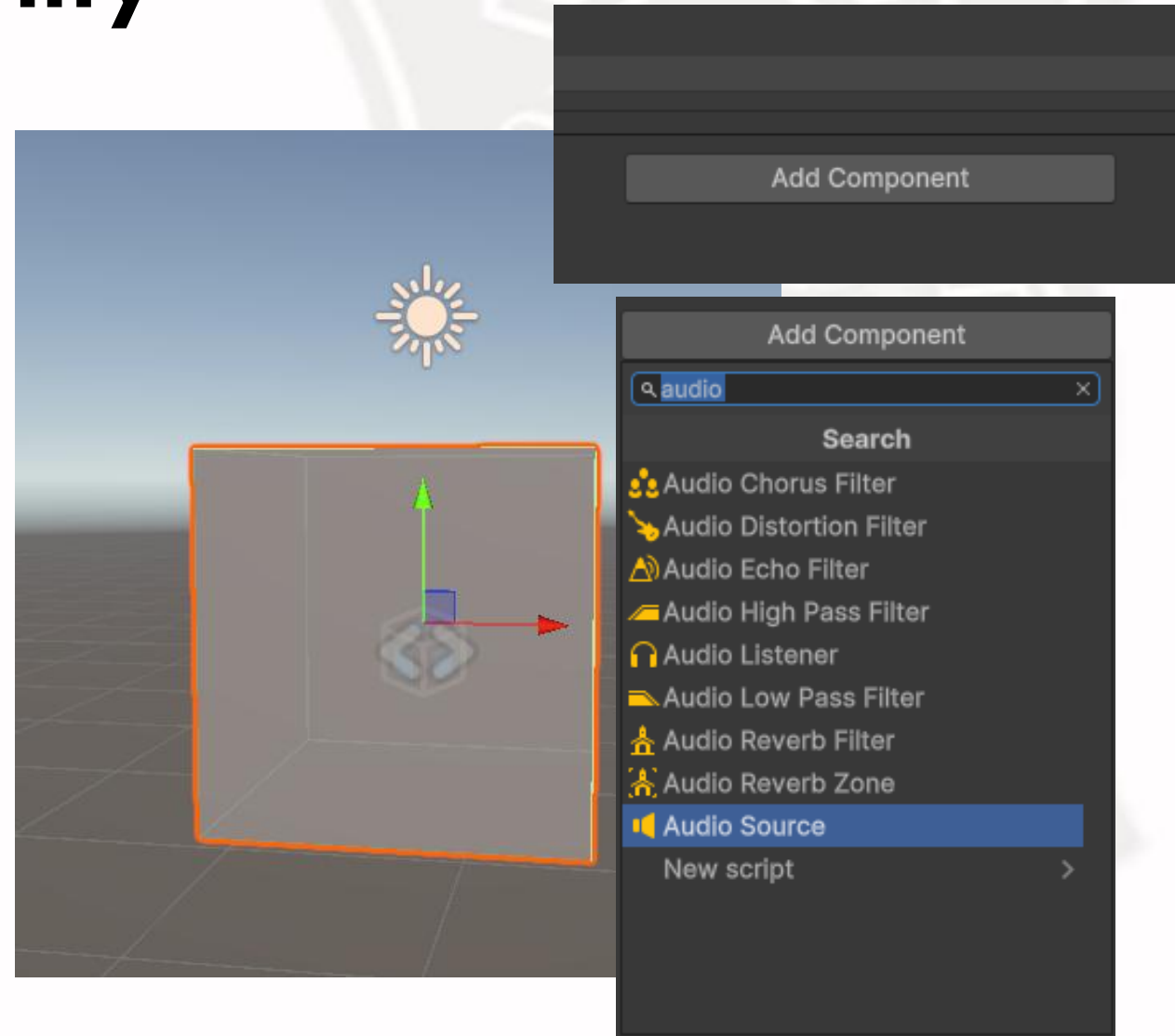
Adding Audio in Unity

- You can open the audio file in the Unity editor and inspect its properties.
- The settings shown on the right will allow you to adjust how the audio is loaded and played to optimize its performance, but we will leave these settings at default for now.



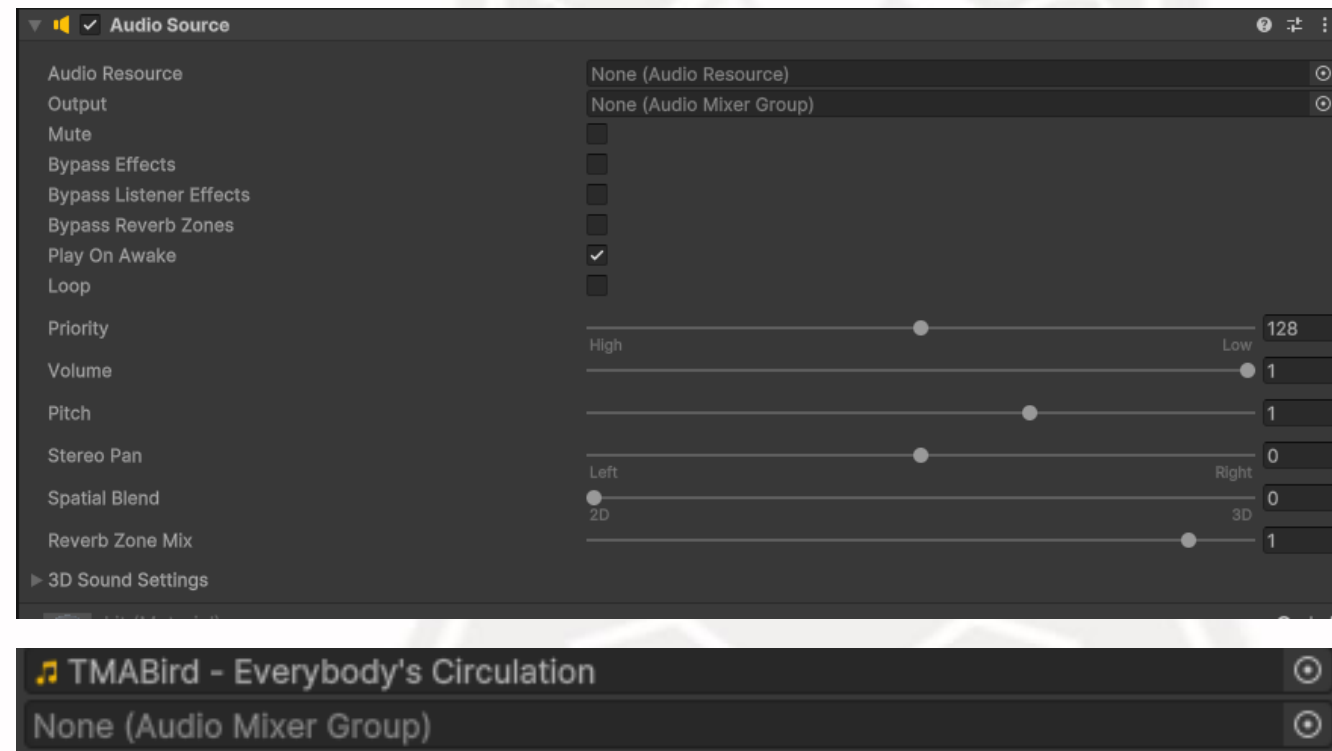
Adding Audio in Unity

- Next, we will need to create an Audio Source which will emit the audio file which we just imported into our project.
- To do this, we select a GameObject and then add a new Audio Source component to it.



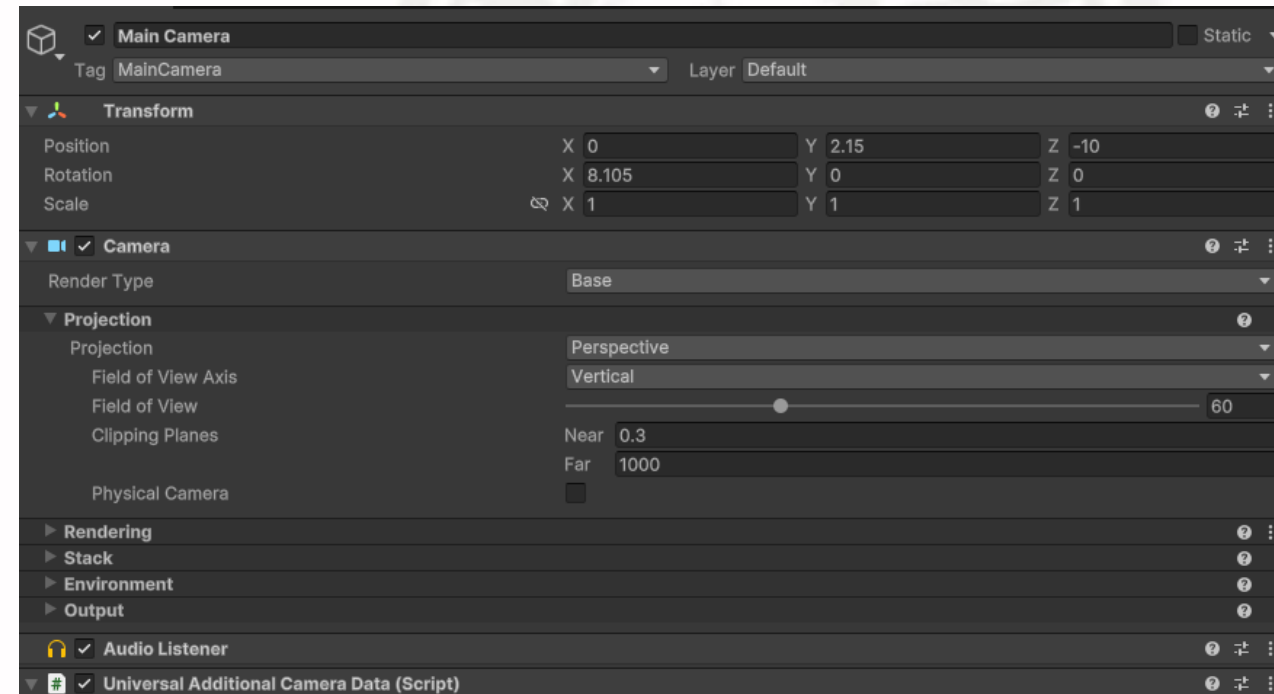
Adding Audio in Unity

- In our Audio source component, we can then add the audio file by selecting **Audio Resource** and then selecting our imported file.



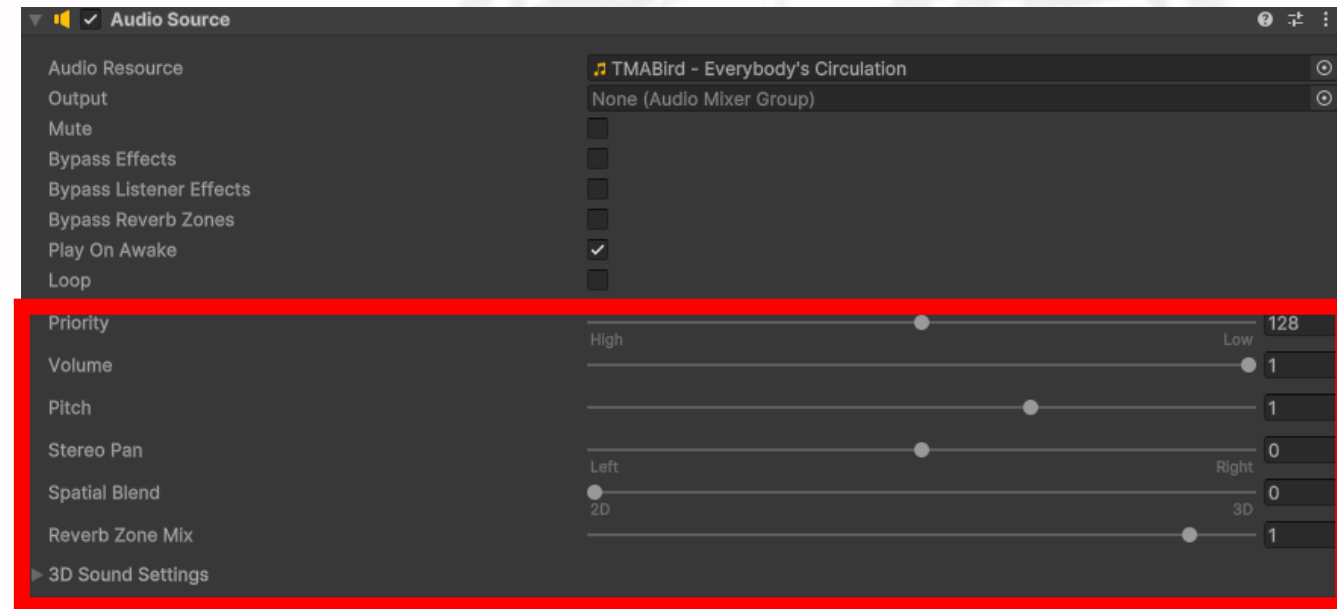
Adding Audio in Unity

- Next, we would typically add an Audio Listener to be able to hear the sound but by default, our main camera already comes with an Audio Listener set up.
- With this, we can hear the audio from the perspective of the camera.



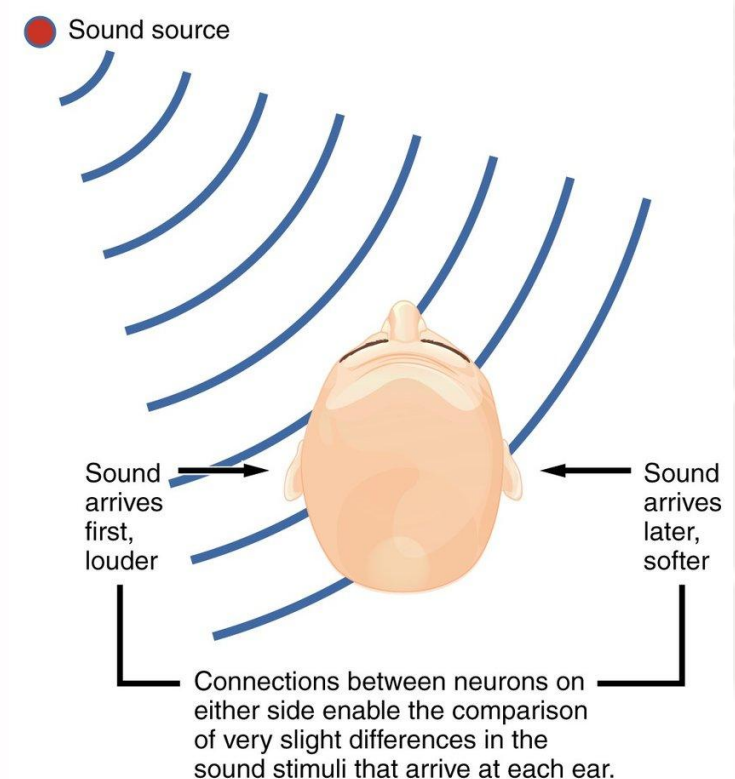
Adding Audio in Unity

- If you play the audio with the default component settings, you might notice that it plays evenly throughout your entire scene at max volume.
- We can adjust this by making changes in the inspector.



Adding Audio in Unity

- Of note is the volume, which changes the overall volume of the sound, and **Spatial Blend**.
- Spatial Blend allows us to create a 3D effect for our audio, depending on how far we adjust the slider.
- The basis for the spatial sound is the Audio Listener.



Adding Audio in Unity

- With this, we can now implement basic sound effects and soundtracks in our game!
- Take note that Audio Source and Audio Listener are components, so they are treated the same way as other components in scripts.

```
AudioSource cubeAudio;

// Start is called once before the first execution
Unity Message | 0 references
void Start()
{
    cubeAudio = GetComponent<AudioSource>();
}

if (Input.GetKeyDown(KeyCode.Space) && audioPlaying == true)
{
    cubeAudio.Pause();
    audioPlaying = false;
}
else if (Input.GetKeyDown(KeyCode.Space) && audioPlaying == false)
{
    cubeAudio.Play();
    audioPlaying = true;
}
```





Audio Mixing

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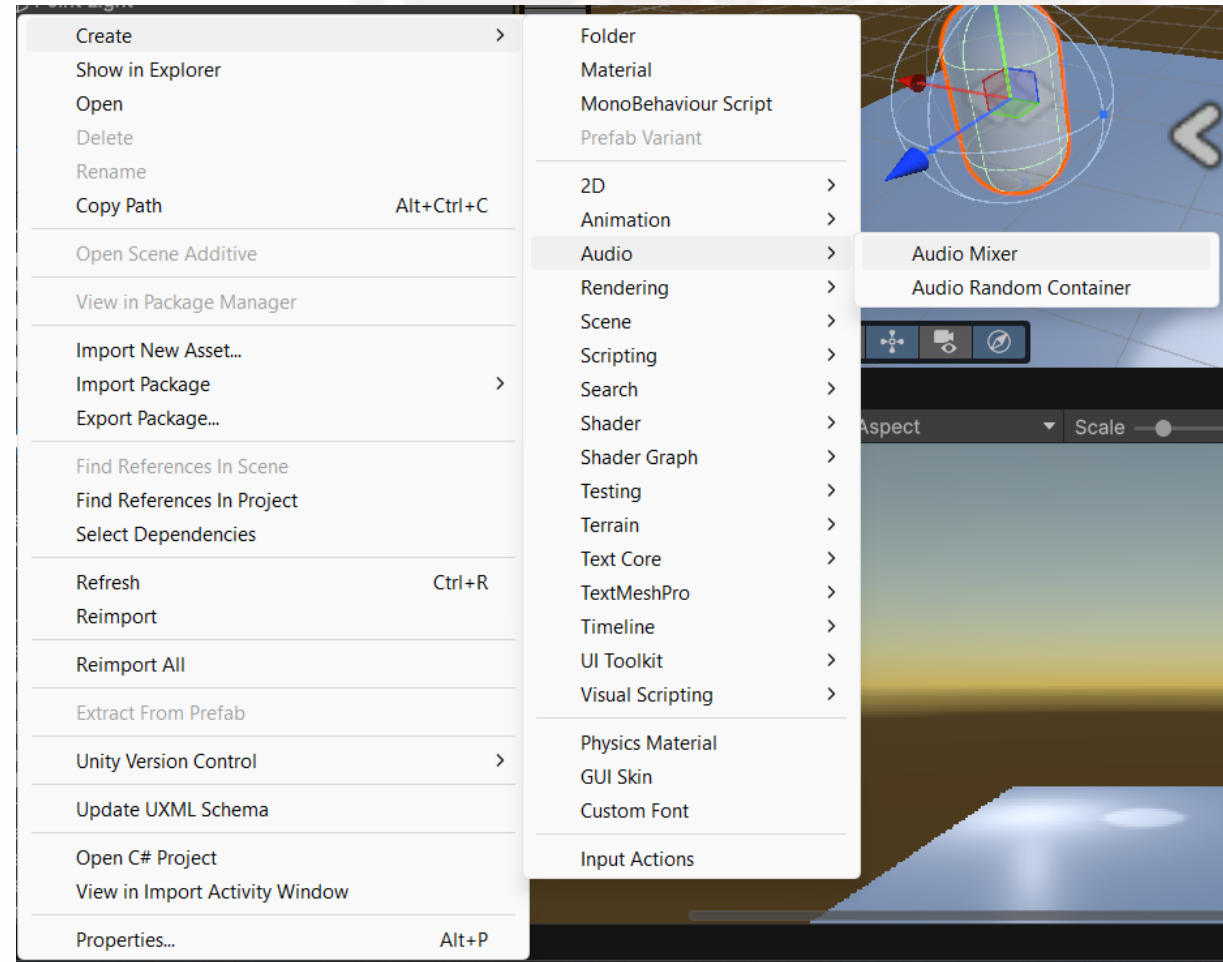
Unity Audio Mixer

- The **Audio Mixer** allows you to mix audio source, apply various effects, and perform audio mastering.
- This is added to our game as a separate asset via the assets folder.
- We can then add it to a GameObject via the Audio Source component.



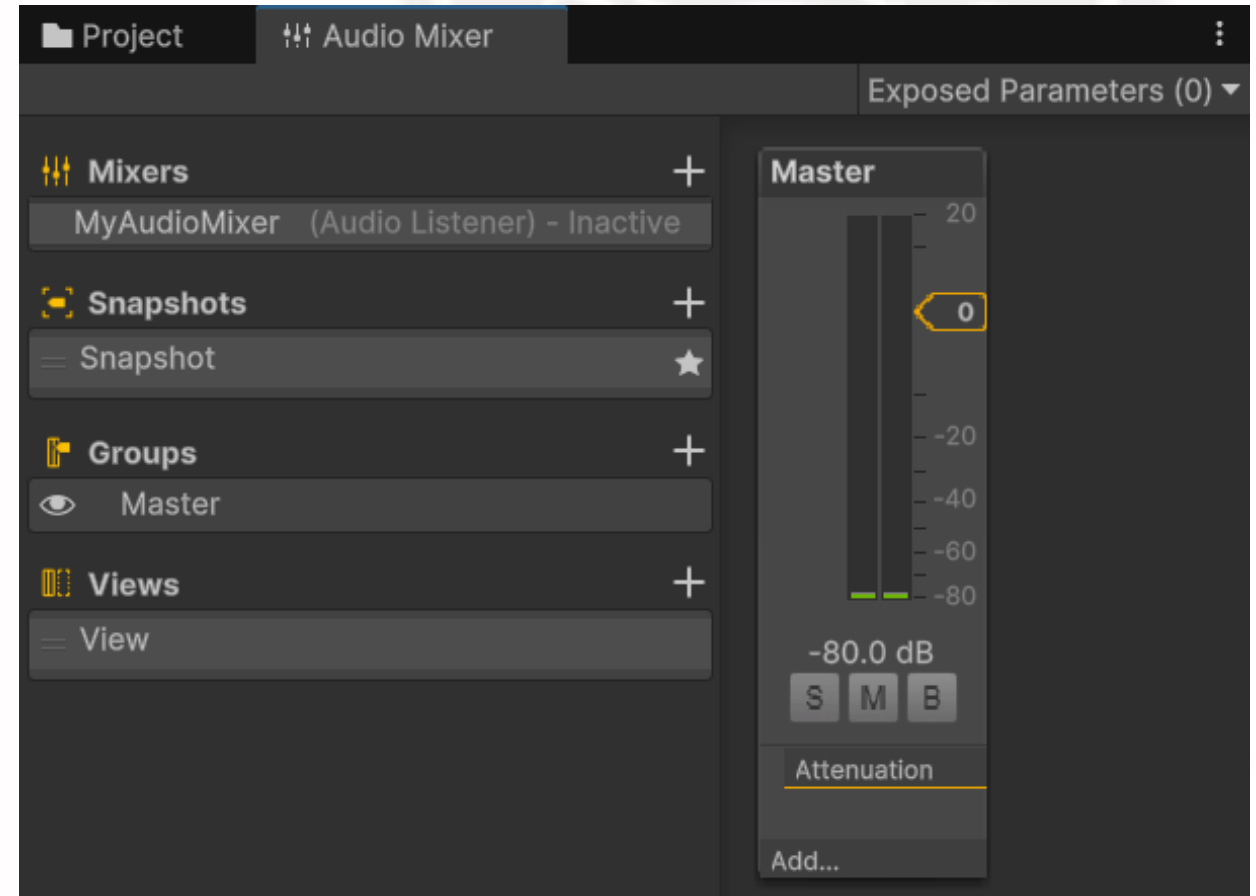
Unity Audio Mixer

- To add an Audio Mixer, we select **Create > Audio > Audio Mixer**.
- This will create a new Audio Mixer asset in our project.
- We can then open the asset by double clicking on it to see its settings.



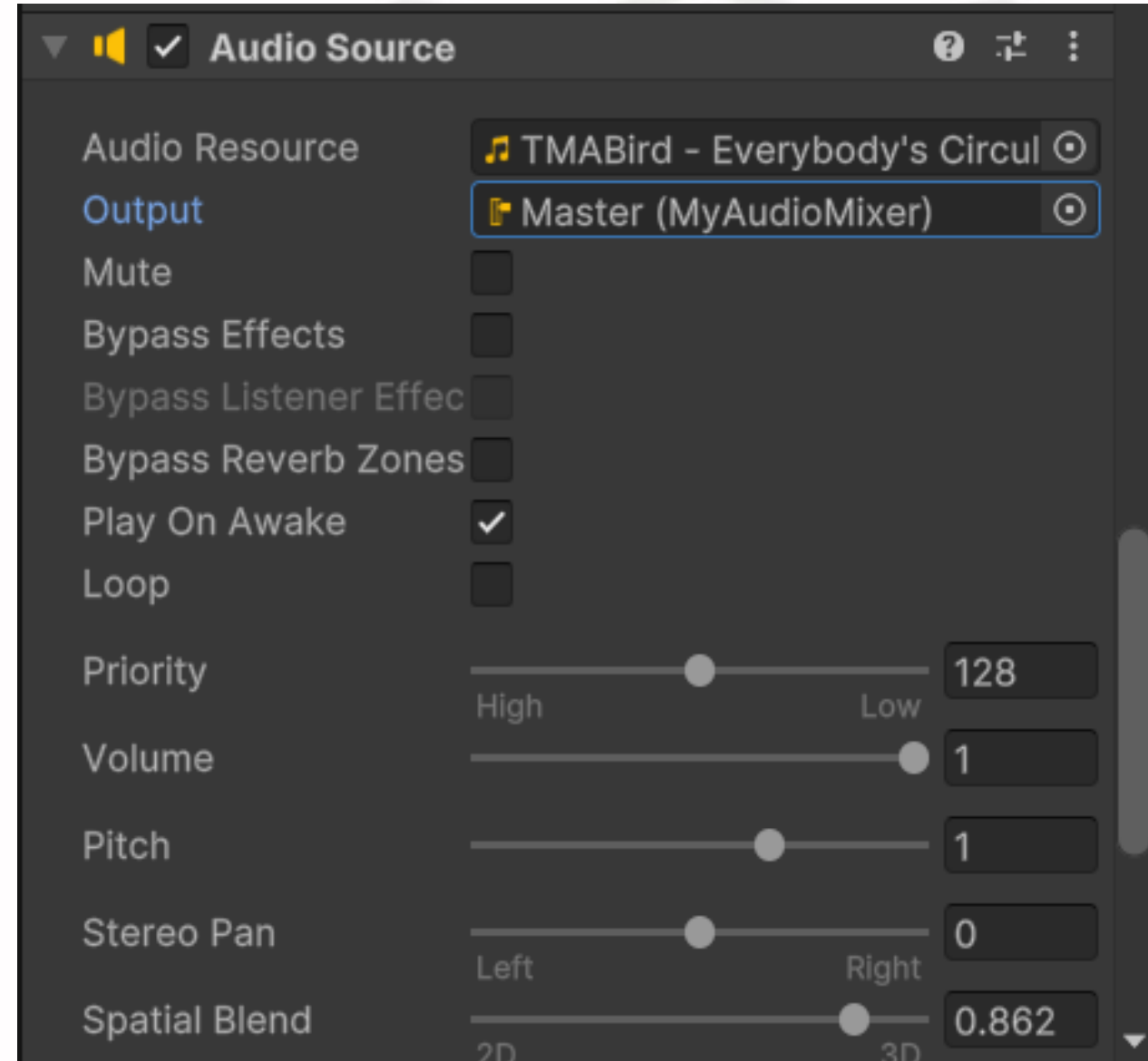
Unity Audio Mixer

- Here, we can adjust the different options for the Audio Mixer, such as adding different effects, or adding different channels.
- In Unity, we call audio channels “groups”.



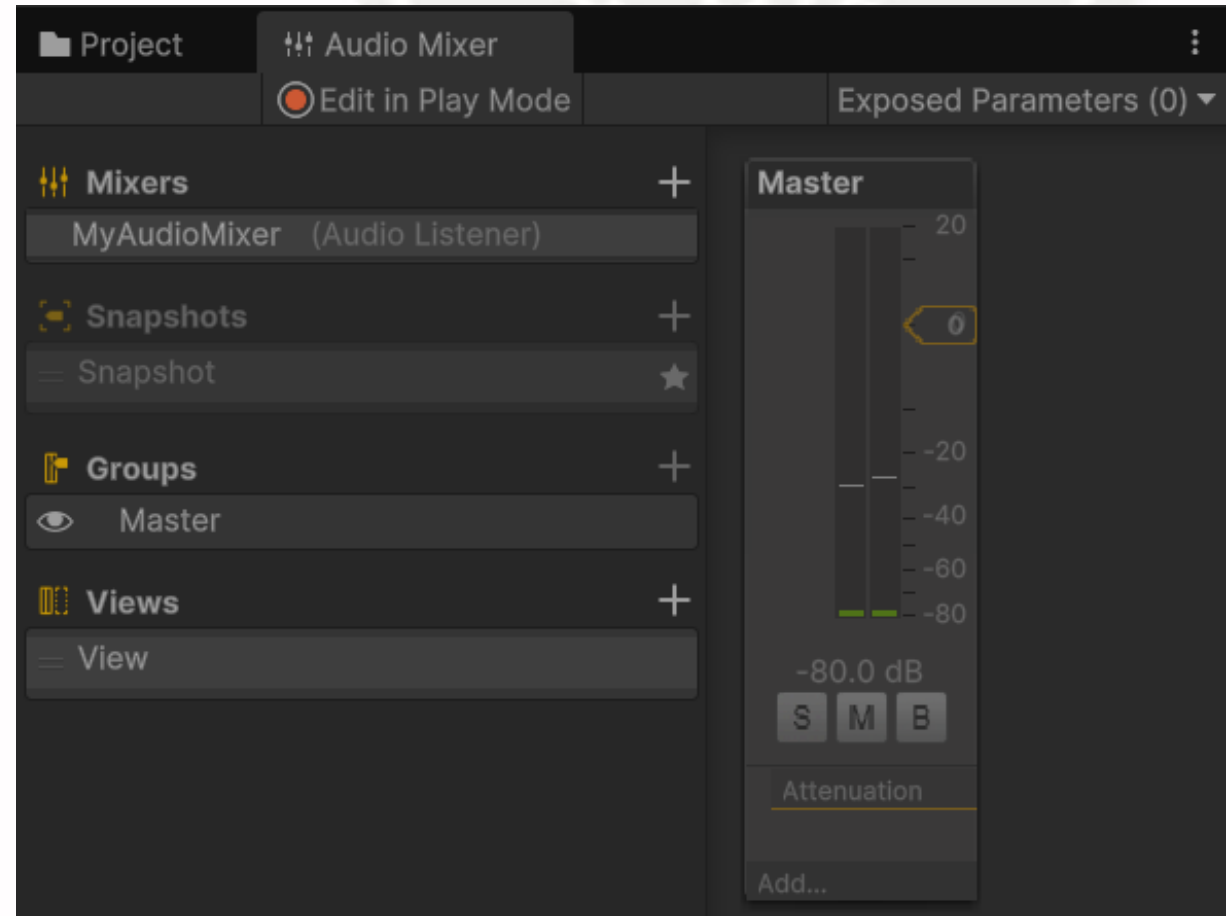
Unity Audio Mixer

- By itself, the Audio Mixer asset does not do anything.
- It needs to be attached to a GameObject for it to function.
- Specifically, we need to add it as part of an Audio Source component.



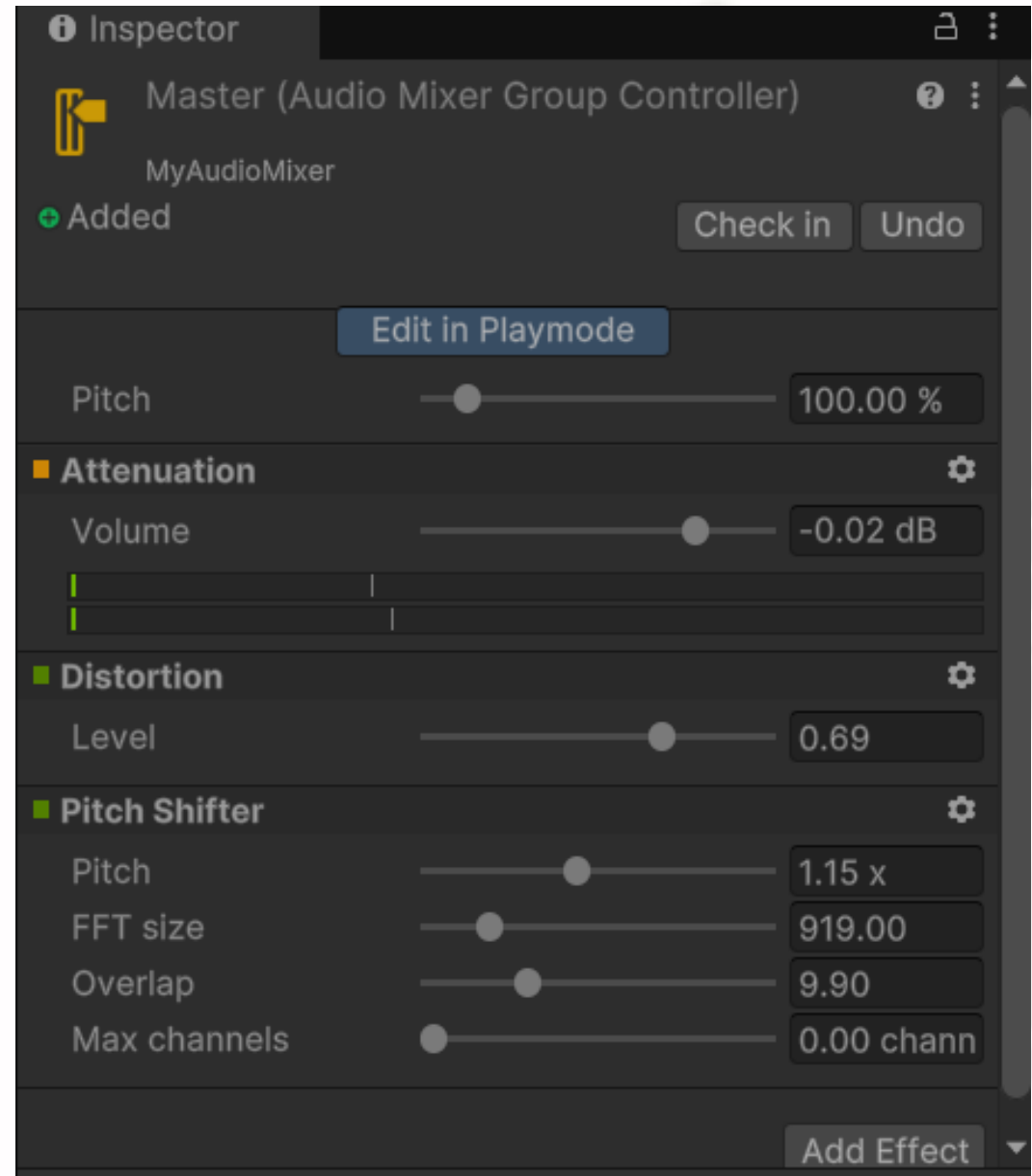
Unity Audio Mixer

- We can adjust the audio in real time through the Audio Mixer while a scene is playing.
- To do this, you simply need to click **Edit in Play Mode** while a scene is running.
- ***Any changes you make to the Audio Mixer while the scene is running are saved.***



Unity Audio Mixer

- You can add multiple effects to your different channels to achieve different effects for your audio files.
- These effects will apply in real time to the audio sources in your scene.





Audio Plug-ins

Unity Audio



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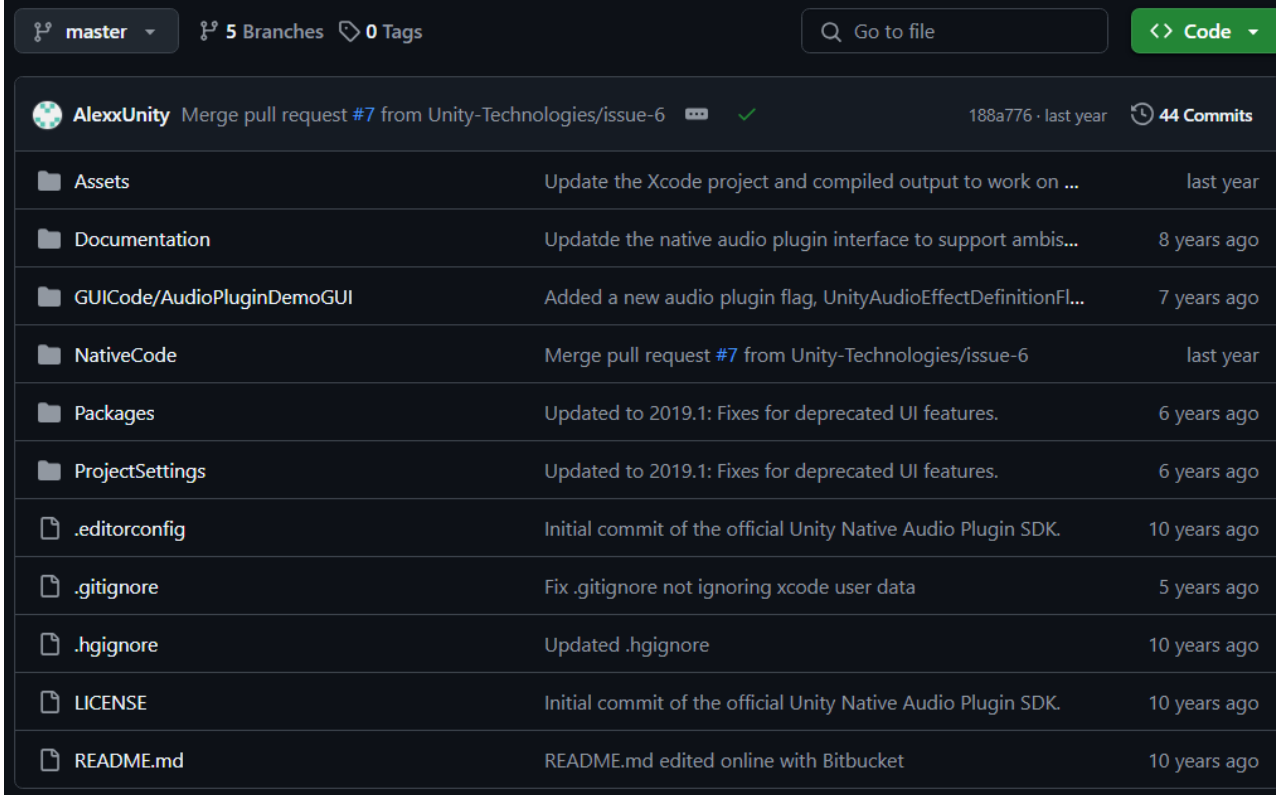
Unity Audio Plug-ins

- These refer to custom audio effects that you can import into your Unity project.
- These are unique audio effects and mixing set ups that are geared towards specific audio themes or genres.
- These can come from the Unity asset store, third party sources, or you can create them yourself.



Unity Audio Plug-ins

- To add Audio Plug-ins, you start by downloading the plugins you would like to use.
- A good starting example is Unity's own official Audio Plugins pack:
<https://github.com/Unity-Technologies/NativeAudioPlugins>



master 5 Branches 0 Tags		
Go to file		
<> Code		
AlexxUnity Merge pull request #7 from Unity-Technologies/issue-6 188a776 · last year 44 Commits		
Assets	Update the Xcode project and compiled output to work on ...	last year
Documentation	Updatde the native audio plugin interface to support ambis...	8 years ago
GUICode/AudioPluginDemoGUI	Added a new audio plugin flag, UnityAudioEffectDefinitionFl...	7 years ago
NativeCode	Merge pull request #7 from Unity-Technologies/issue-6	last year
Packages	Updated to 2019.1: Fixes for deprecated UI features.	6 years ago
ProjectSettings	Updated to 2019.1: Fixes for deprecated UI features.	6 years ago
.editorconfig	Initial commit of the official Unity Native Audio Plugin SDK.	10 years ago
.gitignore	Fix .gitignore not ignoring xcode user data	5 years ago
.hgignore	Updated .hgignore	10 years ago
LICENSE	Initial commit of the official Unity Native Audio Plugin SDK.	10 years ago
README.md	README.md edited online with Bitbucket	10 years ago



Unity Audio Plug-ins

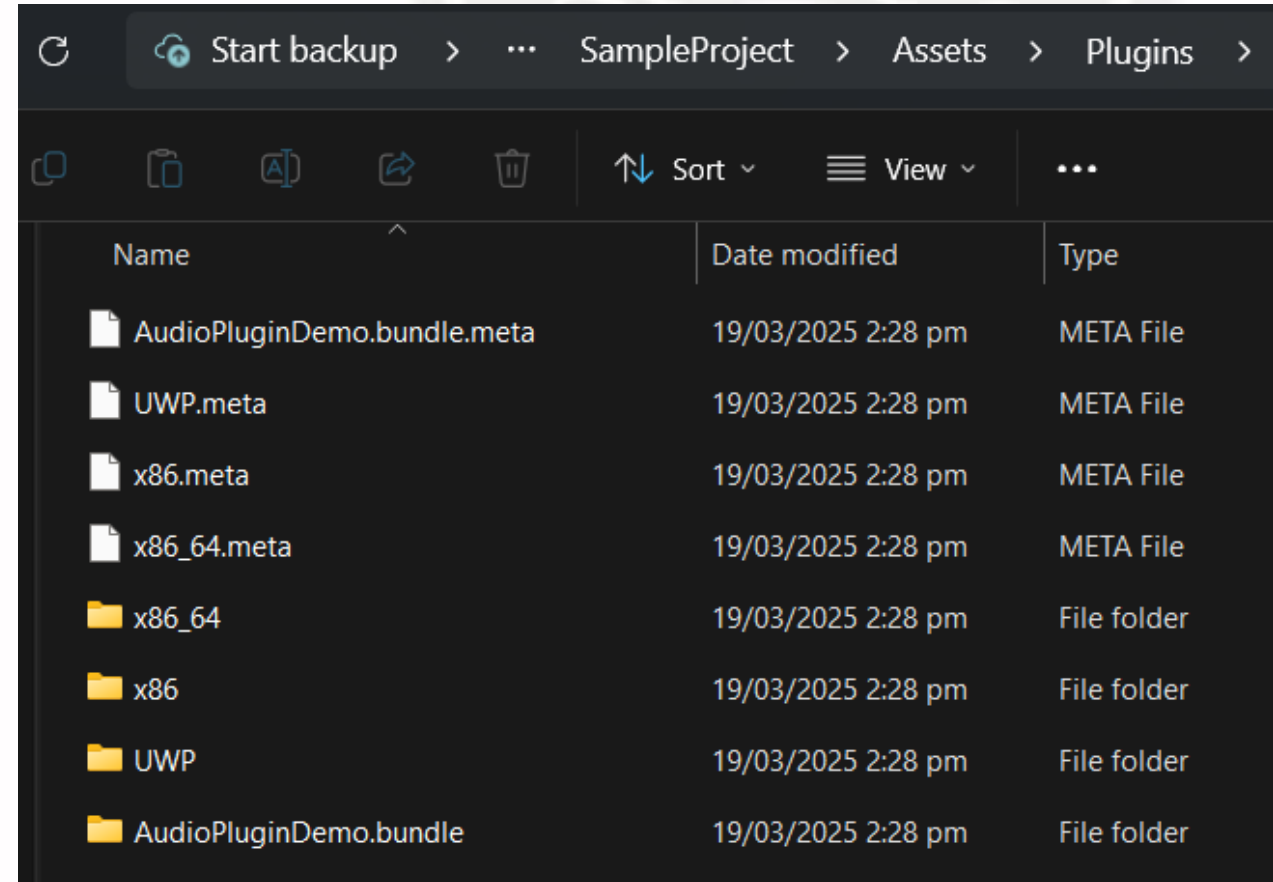
- Once you have your custom plugins downloaded, you simply need to look plugin files themselves.
- You will then create a “Plugins” folder in your own project where you will copy these files to.

Name	Date modified	Type
▼ Today		
AudioPluginDemo.bundle.meta	19/03/2025 2:37 pm	META File
UWP.meta	19/03/2025 2:37 pm	META File
x86.meta	19/03/2025 2:37 pm	META File
x86_64.meta	19/03/2025 2:37 pm	META File
x86	19/03/2025 2:37 pm	File folder
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UWP	19/03/2025 2:37 pm	File folder
AudioPluginDemo.bundle	19/03/2025 2:37 pm	File folder



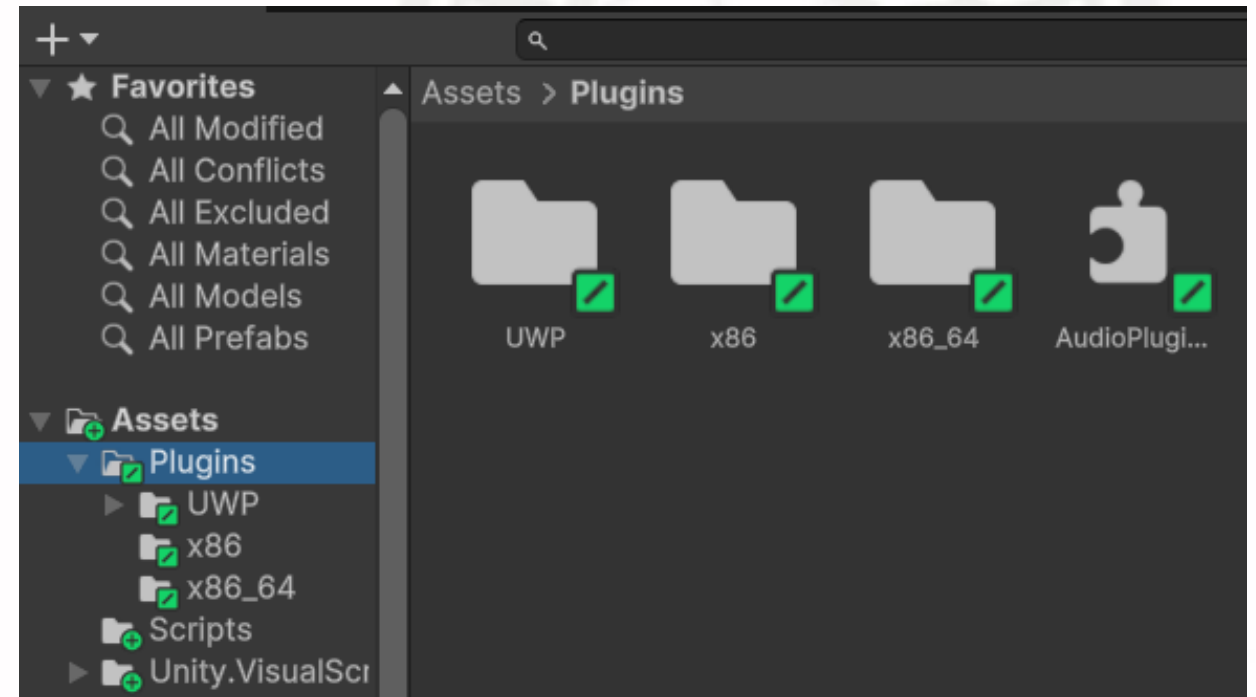
Unity Audio Plug-ins

- After creating a “Plugins” folder in your own project, you can then open your Unity project to load in the plugins.
- Make sure that the plugins you downloaded are compatible with your version of Unity!



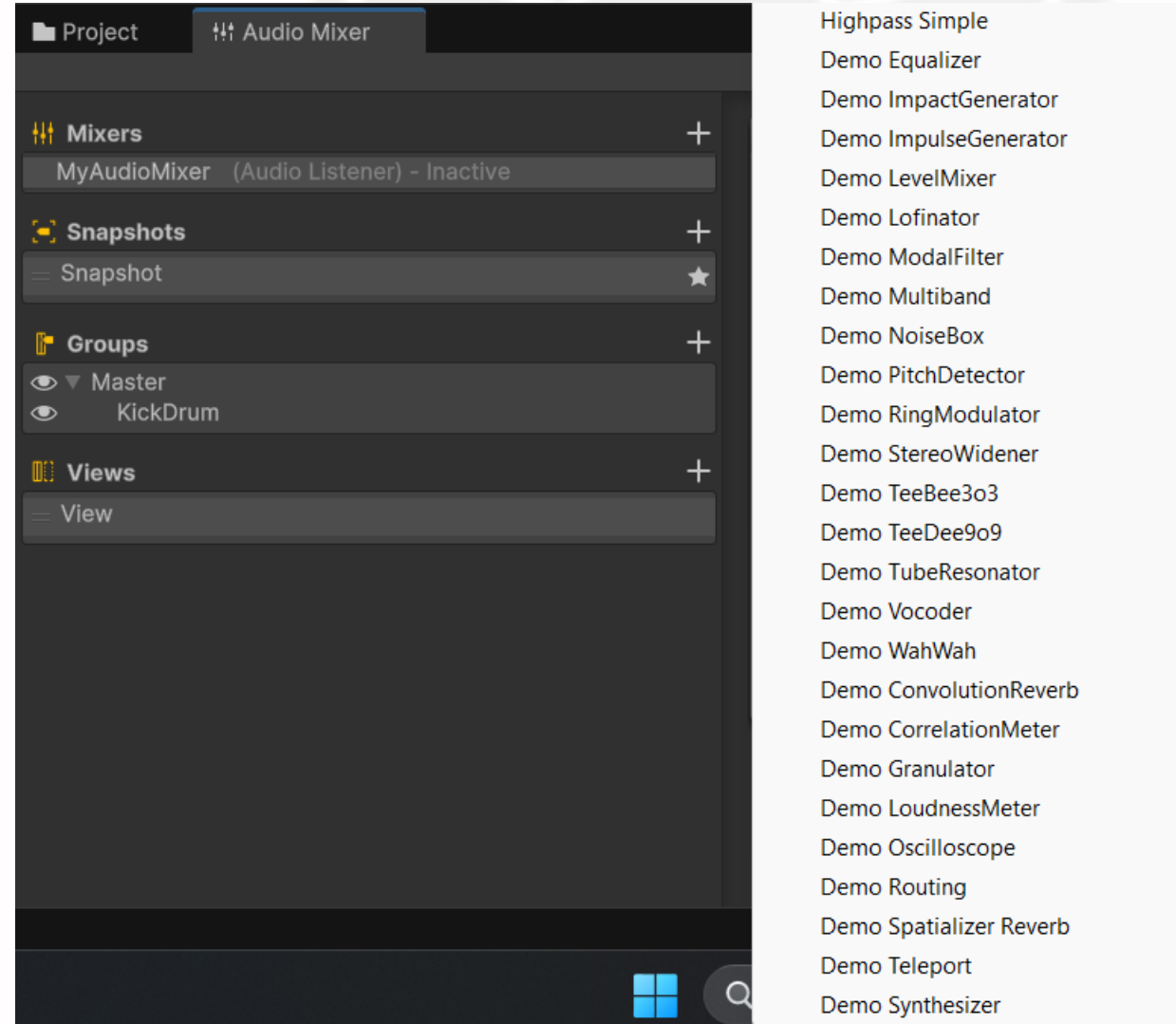
Unity Audio Plug-ins

- Once we have Unity open, we can check if our plugins were successfully imported.
- If they are, we should see them as part of our Assets folder with their files showing up on the Project explorer.



Unity Audio Plug-ins

- Now that we have confirmed that the plugins were correctly imported, we can now add the new audio effects in our Audio Mixer.
- Try experimenting with these new audio effects and see how they work!





Audio Playlists

Unity Audio



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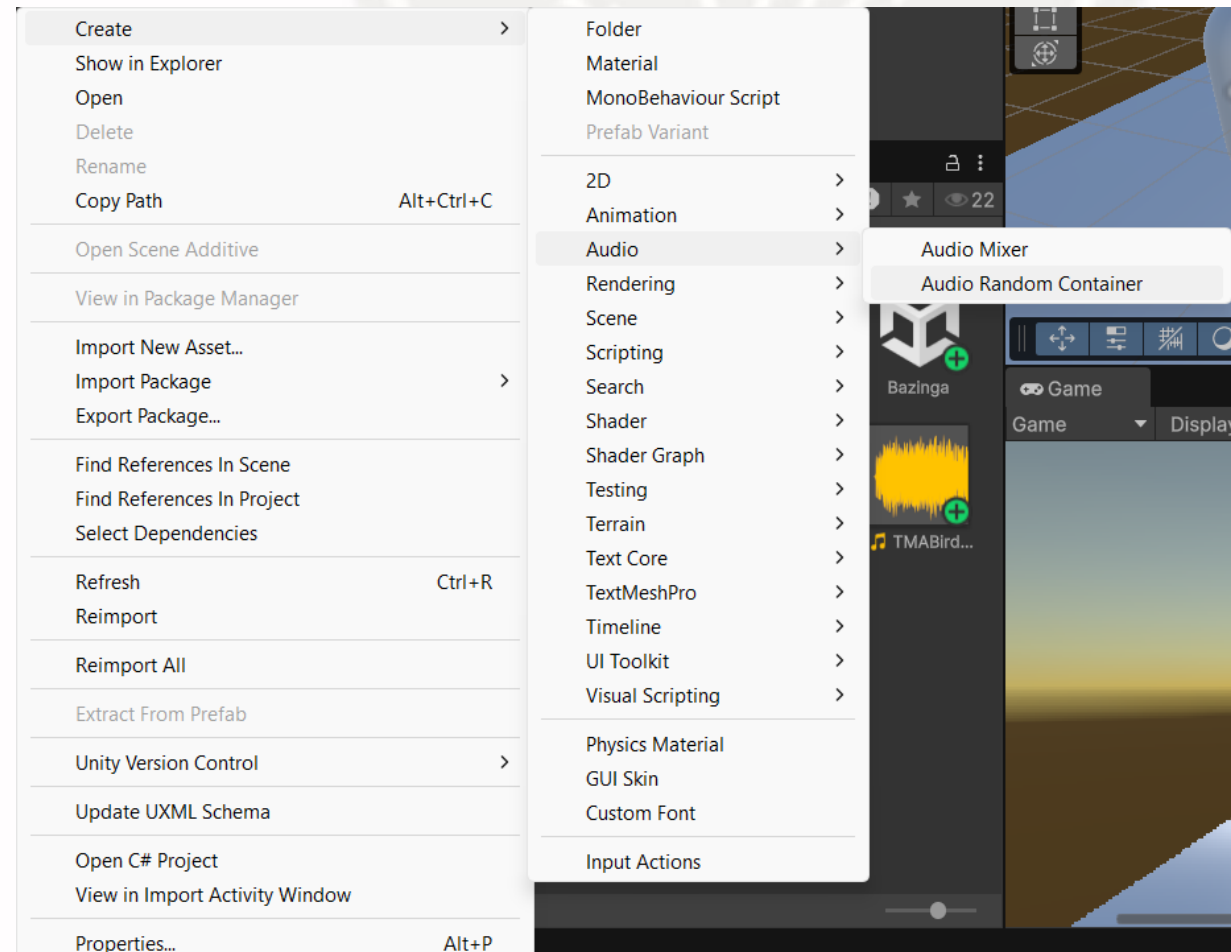
Unity Audio Playlists

- The **Audio Random Container** allows you to set up and play a randomized list of audio clips in your game.
- You can also adjust the volume and adjust effects for the playlist for all audio clips in it.
- This is typically used for repeating sounds such as footsteps, weapons, props, and environmental effects.



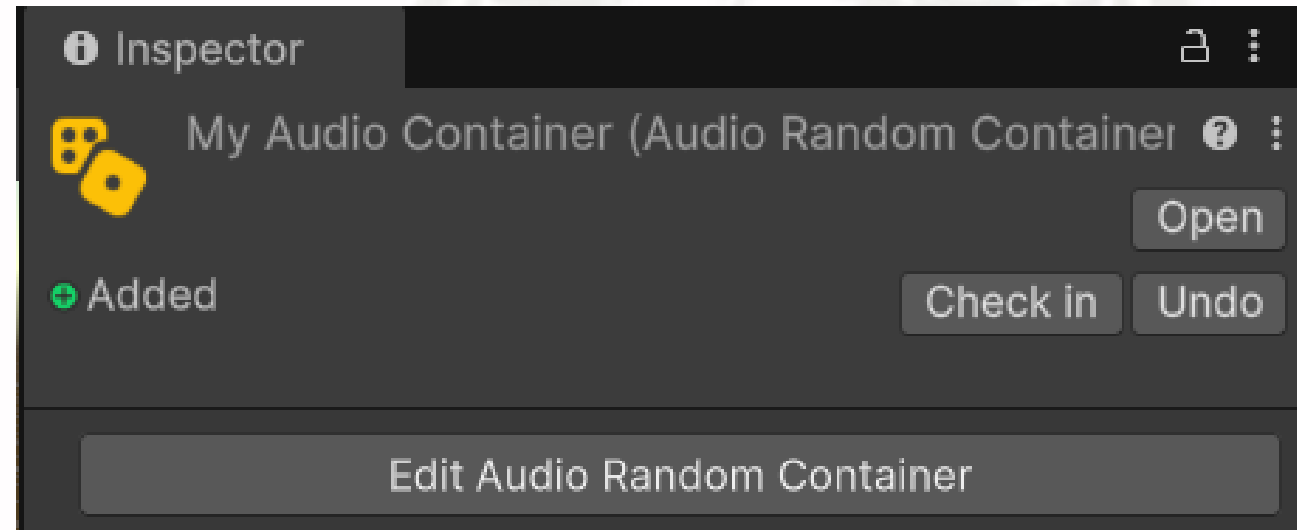
Unity Audio Playlists

- To create an Audio Random Container, we click **Create > Audio > Audio Random Container** to add it as an asset to your game.
- We then double click the newly created asset to add audio files to it.



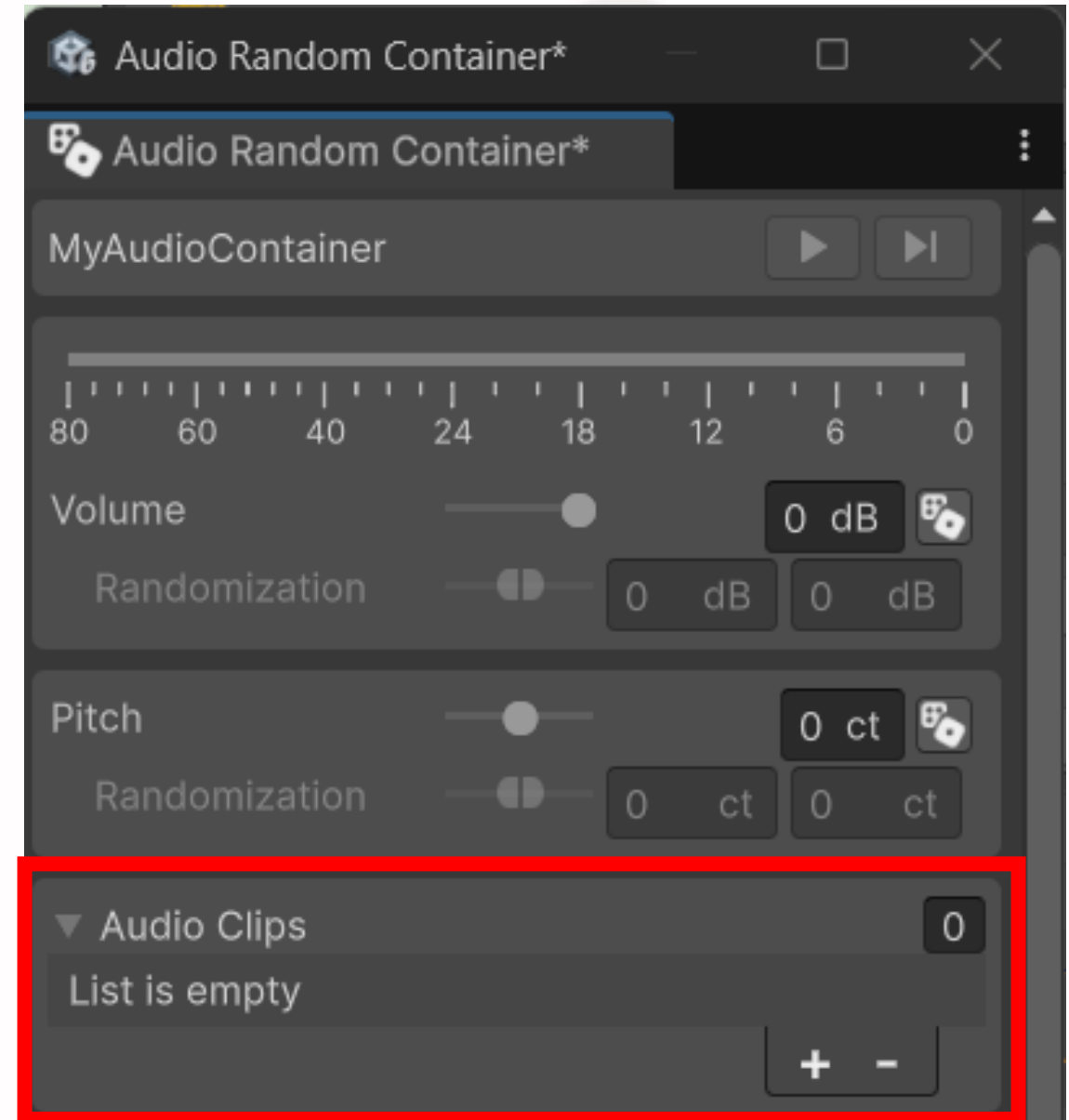
Unity Audio Playlists

- When you open the Audio Random Container asset, you can click **Edit Audio Random Container** to add audio files and edit how they are played in your game.



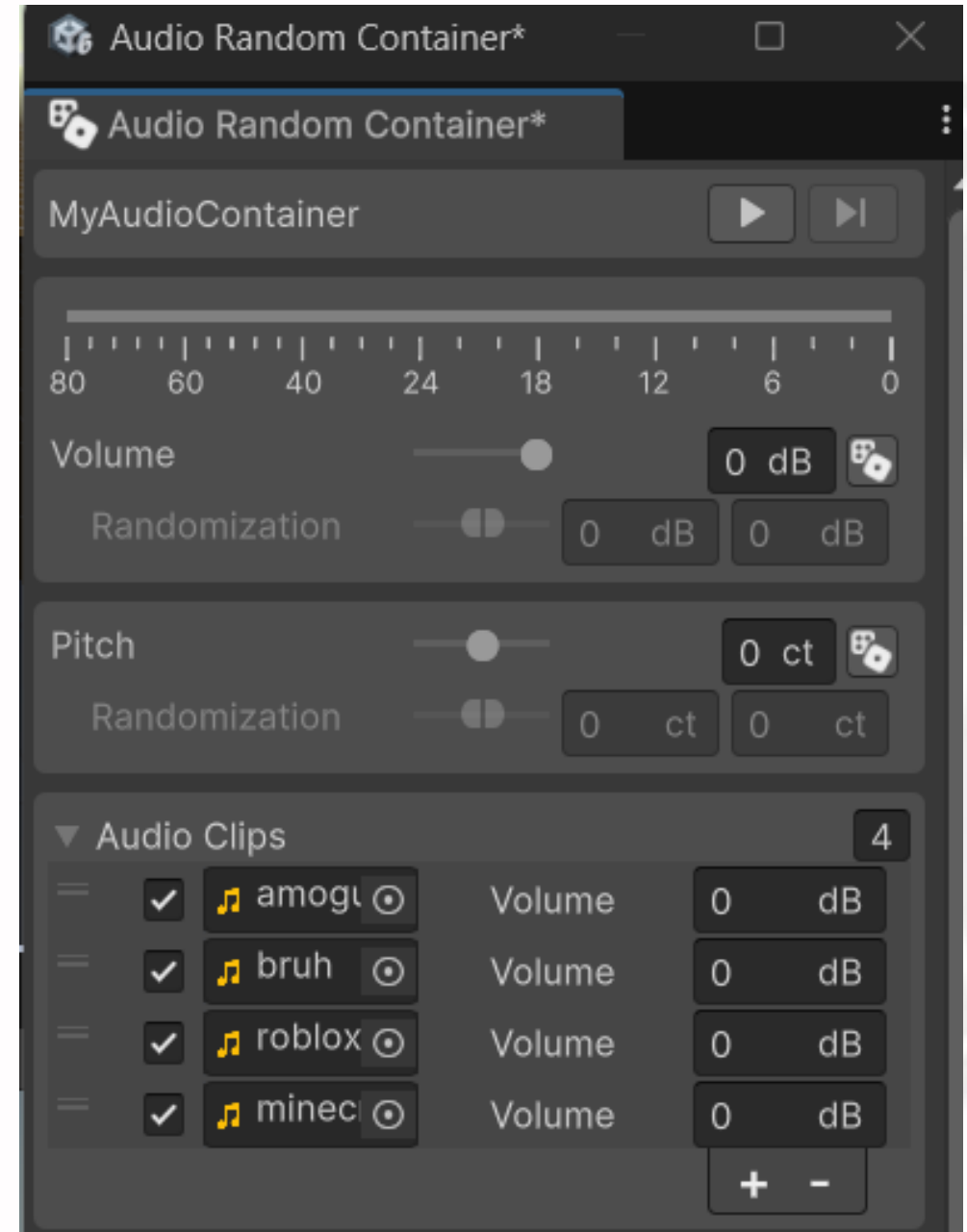
Unity Audio Playlists

- When we open the Audio Random Container asset, we can select the audio clips that we would like to play randomly in our game.
- We can also adjust other options for how the volume and pitch will vary for our audio files.



Unity Audio Playlists

- Once we have added our audio clips, we can now add it as part of our game.
- To add it to our game, we attach it to an Audio Source in one of our Game Objects.



Unity Audio Playlists

- Like individual audio files, we can also adjust the attributes of the random audio files via the Audio Source component or modify how it sounds ever further with a mixer.

